

ARTICLE TYPE

Modeling Exponentiated Log-normal (ELN) Environmental Processes Data with Detection Limits[†]

Kai Peng¹ | Cheng Peng^{*2}

¹School of Data Science and Statistics, Ningbo University of Technology, Zhejiang, China

²Department of Mathematics, West Chester University of Pennsylvania, Pennsylvania, USA

Correspondence

*Corresponding to Cheng Peng. Email: cpeng@wcupa.edu

Present Address

K. Peng: 201 Fenghua Road, Ningbo, Zhejiang, China, 315048

C. Peng: 25 University Ave, West Chester, PA 19383, USA

Summary

In this paper, we propose fitting the exponentiated lognormal (ELN) distribution to environmental data subject to multiple detection limits. As the exponentiated lognormal distribution generalizes the standard lognormal distribution, we provide relevant stochastic and numerical comparisons between the two. Population parameters are estimated using the method of maximum likelihood. Large-sample inference focuses primarily on the population mean and on assessing the necessity of the resilience parameter avoid over-fitting and under-fitting issues in real world applications. Practical guidelines are offered for selecting between asymptotic and bootstrap confidence intervals. Illustrative examples using real-world data are also presented.

KEYWORDS:

Exponentiated lognormal distribution, asymptotic confidence interval, quantile estimation, Pearson χ^2 goodness-of-fit test, detection limits.

1 | INTRODUCTION

Data with non-detect values are common in many areas such as environmental sciences in which instruments cannot detect measurements below certain levels. Various estimation methods have been proposed to handle such data in analysis. Gilliom and Helsel (1986) and Helsel and Gilliom (1986) provide a good review on commonly used existing methods. The recent monograph by Helsel (2005) gives a detailed discussion on various methods available for analyzing data with detection limits.

One-step replacement methods have been proposed and used for decades, but they often yield poor or even invalid results (see Helsel (2006)). Likelihood-based methods have become the primary statistical tools for analyzing such data. Maximum likelihood estimation (MLE) has been employed in this context since the 1950s by researchers including Gupta (1952), Cohen (1957), and Cohen (1959), among others. The monograph by Cohen (1991) provides a systematic introduction to various methods for analyzing left-censored and truncated data. More recently, ? developed asymptotic and bootstrap confidence intervals for population means and variances based on left-censored data with multiple detection limits, and proposed using the bootstrap sampling distribution to assess the reliability of the asymptotic intervals.

The *Expectation-Maximization* (EM) algorithm, proposed by Dempster, Laird, and Rubin (1977), is an iterative procedure for finding the maximum likelihood estimates (MLEs) of model parameters. The core idea of the EM algorithm is to estimate the model parameters using the completely observed data, then use these estimated parameters to predict the censored values, and repeat this process until convergence is achieved. In other words, the EM algorithm can be viewed as a model-based iterative replacement method for obtaining MLEs. This insightful perspective was also discussed in El-Shaarawi and Esterby (1992).

Some nonparametric approaches have also been discussed in literature. Reader are referred to, for example, Helsel (1990), Gilliom and Helsel (1986), El-Shaarawi (1989), and Singh and Nocerino (2002).

[†]This is an example for title footnote.

Since environmental data are positive and typically skewed, the parametric distributions commonly used for non-detect data include the lognormal, gamma, and Weibull distributions (see Zhao and Frey (2006)). Among these, the lognormal distribution is the most widely applied in practice, largely because lognormal data can be transformed to achieve normality, facilitating the use of methods based on the normal distribution. In this paper, we propose fitting the exponentiated lognormal distribution to skewed environmental data. The exponentiated lognormal distribution includes the standard lognormal distribution as a special case. Its cumulative distribution function (CDF) is obtained by raising the CDF of the lognormal distribution to a power. That is,

$$G(x) = \left[\Phi \left(\frac{\log(x) - \xi}{\eta} \right) \right]^\alpha, \quad (1)$$

where α is an additional parameter introduced into the cumulative distribution function (CDF) of the lognormal distribution, which has location parameter ξ and scale parameter η . Here, $\Phi(\cdot)$ denotes the CDF of the standard normal distribution. This distribution belongs to a broader family of distributions discussed in Marshall and Olkin (2007). The parameter α is referred to as the resilience parameter, also known as the proportional reverse hazard rate in lifetime and survival analysis. Martínez-Flórez, Bolfarine, and Gómez (2014) termed this model the log-power-normal distribution and applied it to fit air pollution data.

In this study, we adopt this model to analyze environmental data subject to multiple detection limits. Multiple detection limits in environmental studies arise from a variety of practical sources, including changes in laboratory instrumentation or analytical methods over time, differences in detection capabilities across laboratories, and variations in sample matrices or preparation techniques. They also result from compound-specific analytical properties, regulatory reporting requirements that differ from instrumental detection limits, and the compilation of historical data from multiple studies or monitoring programs with differing analytical protocols. These diverse sources create datasets where censoring thresholds are not uniform, requiring rigorous statistical methods that can accommodate varying detection limits while properly handling the skewed nature of environmental measurements.

The remainder of this paper is organized as follows. In Section 2, we briefly present some results comparing the lognormal and exponentiated lognormal distributions. Section 3 describes the maximum likelihood estimation of the model parameters and discusses the asymptotic properties of the estimated population parameters. Section 4 is devoted to constructing confidence intervals for the population mean and quantiles of the exponentiated lognormal distribution. Goodness-of-fit tests for assessing correct model specification are discussed in Section 5. In Section 6, we conduct simulation studies to evaluate the power of the proposed testing procedures. Several illustrative examples using real-world environmental datasets, along with practical guidelines for selecting appropriate confidence intervals, are presented in Section 7. Finally, discussions and concluding remarks are provided in Section 8.

2 | PROPERTIES AND VISUAL ANALYSES OF ELN

We first introduce some notations and definitions related to stochastic orders. Let the cumulative distribution function (CDF) and probability density function (PDF) of the lognormal distribution be denoted as

$$F(x) = \Phi \left(\frac{\log(x) - \xi}{\eta} \right) \quad \text{and} \quad f(x) = \frac{1}{x\eta} \phi \left(\frac{\log(x) - \xi}{\eta} \right) \quad (2)$$

where ξ and η are location and scale parameters respectively. We also denote the density function of the exponentiated lognormal distribution to be

$$g(x) = \frac{\alpha}{x\eta} \phi \left(\frac{\log(x) - \xi}{\eta} \right) \left[\Phi \left(\frac{\log(x) - \xi}{\eta} \right) \right]^{\alpha-1} \quad (3)$$

We observe several patterns from Figure 1. First, all density curves are skewed to the right (positively skewed), regardless of the choice of parameter values. In addition:

As the resilience parameter increases, the corresponding density curve becomes flatter for fixed ξ and η . This implies that both the mean and variance increase as the resilience parameter increases.

When $\alpha < 1$, both the mean and variance of the exponentiated lognormal distribution are smaller than those of the lognormal distribution. In contrast, when $\alpha > 1$, both the mean and variance of the exponentiated lognormal distribution exceed those of the lognormal distribution.

Suppose X has cumulative distribution function $F(x) = \Phi[(\log x - \xi)/\eta]$ and Y has cumulative distribution function $G(x)$. Then the relationship between X and Y is given by the following result.

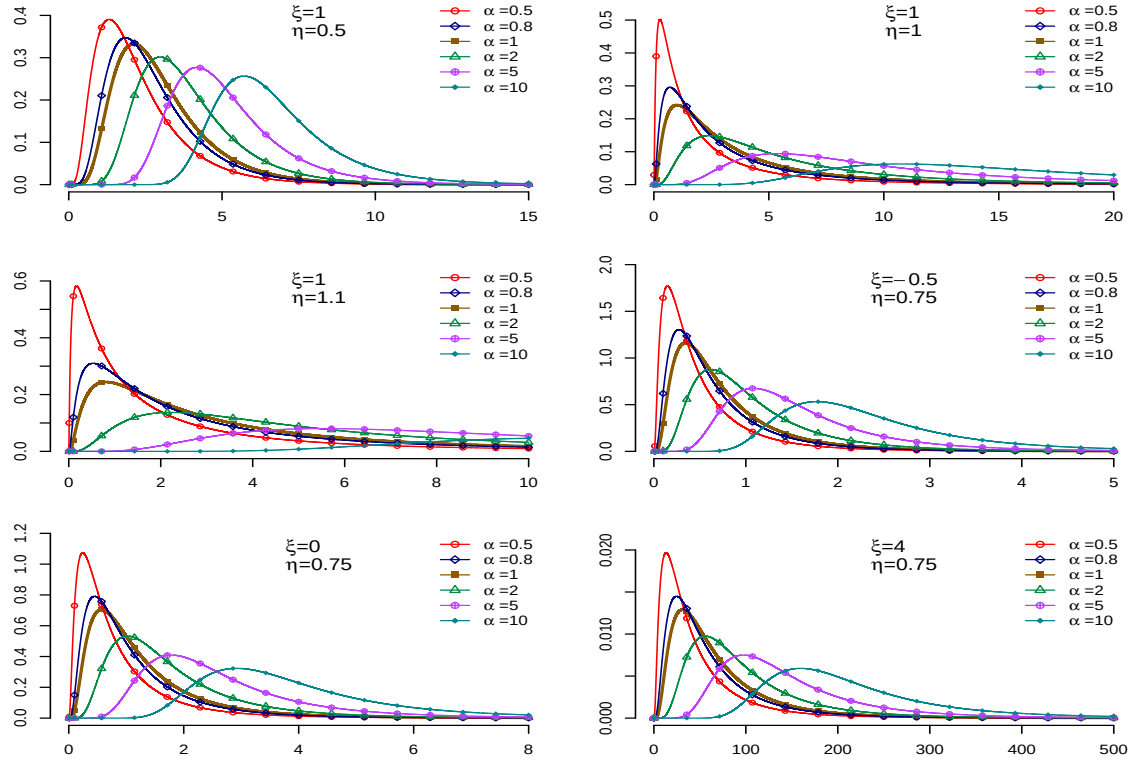


FIGURE 1 Density curves of exponentiated lognormal distributions with various values of resilience parameter α , the location and scale parameters of the reference lognormal distribution ξ and η , respectively.

Theorem 1. Let Y and X be two random variables having distributions in (1) and (2). We have $P(Y > X) = \alpha/(1 + \alpha)$.

Proof. We use ROC curve method to show this result. Let $\bar{G}(x)$ and $\bar{F}(x)$ be the survival function of Y and X . From chapter 4 of ?, the ROC curve is defined to be $[\bar{F}(x), \bar{G}(x)]$ for all $x \in \mathbb{R}^+$. Let $t = \bar{F}(x) = 1 - \Phi[(\log(x) - \xi)/\eta]$. Then $\Phi^{-1}(1 - t) = (\log(x) - \xi)/\eta$. The ROC curve is then defined as

$$\text{ROC}(t) = 1 - \left[\Phi \left(\frac{\log(x) - \xi}{\eta} \right) \right]^\alpha = 1 - [\Phi(\Phi^{-1}(1 - t))]^\alpha = 1 - (1 - t)^\alpha. \quad (4)$$

From Result 4.2 of Pepe (2004) [page 69], we have

$$P(Y > X) = \int_0^1 \text{ROC}(t) dt = \int_0^1 [1 - (1 - t)^\alpha] dt = \frac{\alpha}{1 + \alpha}. \quad (5)$$

The proof is complete. $\square$

The key numerical characteristics of interest in this paper are the inference about the expectation and variance of skewed population that follows the exponentiated lognormal distribution. Both expectation and variance are related to moments. The following result concerns the monotonicity of the r -th moment of Y . Note that the r -th moment of Y is defined as

$$E[Y^r] = \int_0^\infty y^r d \left[\Phi \left(\frac{\log(y) - \xi}{\eta} \right) \right]^\alpha. \quad (6)$$

After changing variable

$$x = \left[\Phi \left(\frac{\log(y) - \xi}{\eta} \right) \right]^\alpha, \quad \text{that is, } y = \exp \left[\xi + \eta \Phi^{-1}(x^{\frac{1}{\alpha}}) \right].$$

We re-express the r -th moment as

$$E[Y^r] = \int_0^1 \exp \left\{ r \left[\xi + \eta \Phi^{-1}(x^{\frac{1}{\alpha}}) \right] \right\} dx. \quad (7)$$

Let $t = x^{1/\alpha}$, then $dx = \alpha t^{\alpha-1} dt$. With this change of variable, we re-express the t -th moment in the following

$$\mu_Y^{(r)} = \int_0^1 \alpha t^{\alpha-1} \exp[r\xi + r\eta\Phi^{-1}(t)] dt \quad (8)$$

Therefore, the mean and variance of the exponentiated lognormal distribution are given respectively by

$$\mu = E[Y] = \int_0^1 \alpha t^{\alpha-1} \exp \left[\xi + \eta \Phi^{-1}(t) \right] dt \quad (9)$$

and

$$\sigma^2 = V(Y) = \int_0^1 \alpha t^{\alpha-1} \exp \left\{ 2 \left[\xi + \eta \Phi^{-1}(t) \right] \right\} dt - \mu^2 \quad (10)$$

To highlight the functional relationship between the moment and the resilience parameter α , we denote $\mu_Y^{(r)}(\alpha) = E[Y^r]$.

Theorem 2. Under model (1), the r -th moment of the exponentiated lognormal distribution is an increasing function in α for fixed ξ and η .

Proof. We only show the first order derivative is positive for all $\alpha \in \mathbb{R}^+$. Observe that

$$\frac{d}{d\alpha} \mu_Y^{(r)}(\alpha) = \int_0^1 \exp \left\{ r \left[\xi + \eta \Phi^{-1}(x^{\frac{1}{\alpha}}) \right] \right\} \cdot r\eta \frac{d\Phi^{-1}(x^{1/\alpha})}{d\alpha} dx. \quad (11)$$

Denote $\psi = \Phi^{-1}(x^{1/\alpha})$. We solve for x and obtain $[\Phi(\psi)]^\alpha = x$, which is equivalent to $\alpha \log[\Phi(\psi)] = \log(x)$. Taking derivative of implicit function, we have

$$\frac{d\psi}{d\alpha} = \frac{-x^{1/\alpha} \log x^{1/\alpha}}{\alpha \phi[\Phi^{-1}(x^{1/\alpha})]} > 0 \quad (12)$$

since $0 < x < 1$, $\log(x) < 0$. Substituting $d\psi/d\alpha$ in (12) to (11) and notice that the integrand in (11) is positive. We conclude that $\frac{d}{d\alpha} \mu_Y^{(r)}(\alpha) > 0$. Therefore, the r -th moment is increasing in α . $\square$

Corollary 1. Given fixed location and scale parameters in the exponentiated lognormal distribution, the expectation is a monotonically increasing function of the resilience parameter α .

Proof. The monotonicity of the expectation was established in Theorem 2. We now show that the variance of the exponentiated lognormal distribution is also increasing in α . In the proof of Theorem 2, we demonstrated that $\psi = \Phi^{-1}(x^{1/\alpha})$ is increasing in α . Similar arguments can be used to show that ψ is also increasing in x for $x \in [0, 1]$. $\square$

Instead of presenting cumbersome derivations, we use the following figure to visualize how the parameter values affect the shape of the variance.

Figure 2 shows that, when any two of the three parameters are held fixed, the variance is an increasing function of the remaining parameter. This observation is consistent with the patterns seen in Figure 1. We also note from Figure 1 that exponentiated lognormal distributions are positively skewed. Figure 3 below confirms this positive skewness and provides additional insight into how each parameter affects the degree of skewness.

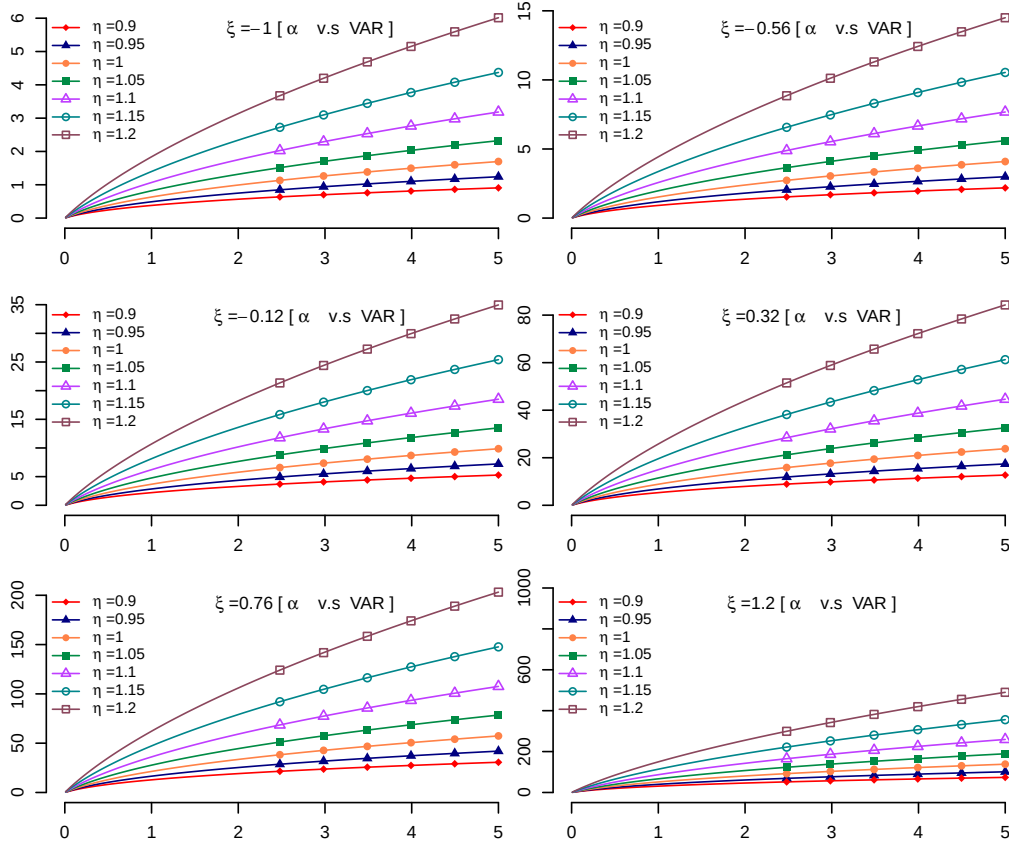


FIGURE 2 Variances of exponentiated lognormal distributions with various values of resilience parameter α , the location and scale parameters of the reference lognormal distribution ξ and η , respectively.

Figure 3 shows that the coefficient of skewness is always positive (the parameter values were chosen to provide an overall view while avoiding overflow errors and divergence issues in the computations). As seen in the top three panels, for fixed ξ and η , the skewness coefficient is a decreasing function of α , which is consistent with the pattern observed in Figure 1. In contrast, for fixed α and ξ , the coefficient of skewness increases with η . The parameter ξ , however, does not appear to have a significant impact on the degree of skewness.

3 | MAXIMUM LIKELIHOOD ESTIMATION AND ASYMPTOTIC RESULTS

Let $y_1, \dots, y_n$ be a Type I left-censored sample drawn from a normal population $N(\xi, \eta^2)$, where $y_i = (x_i, c_i)$ and c_i is the censoring indicator such that $c_i = 1$ if x_i is censored and $c_i = 0$ if x_i is fully observed. From (1) and (3), the likelihood function based on the left-censored data can be written as

$$L(\xi, \eta, \alpha) = \prod_{i=1}^n \left[\Phi \left(\frac{\log(x_i) - \xi}{\eta} \right) \right]^{\alpha c_i} \left\{ \frac{\alpha}{x_i \eta} \phi \left(\frac{\log(x_i) - \xi}{\eta} \right) \left[\Phi \left(\frac{\log(x_i) - \xi}{\eta} \right) \right]^{\alpha-1} \right\}^{1-c_i}, \quad (13)$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ denote the cumulative distribution function and probability density function of the standard normal distribution, respectively. The parameters ξ , η , and α represent the location, scale, and shape parameters of the underlying exponentiated lognormal population.

Since Type I left-censored samples involve fixed censoring limits, we simplify the notation by letting $x_1, x_2, \dots, x_{n_0}$ denote the set of completely observed values. The multiple detection limits are given by $(d_1, d_2, \dots, d_k)$, with corresponding censoring counts $m_1, m_2, \dots, m_k$.

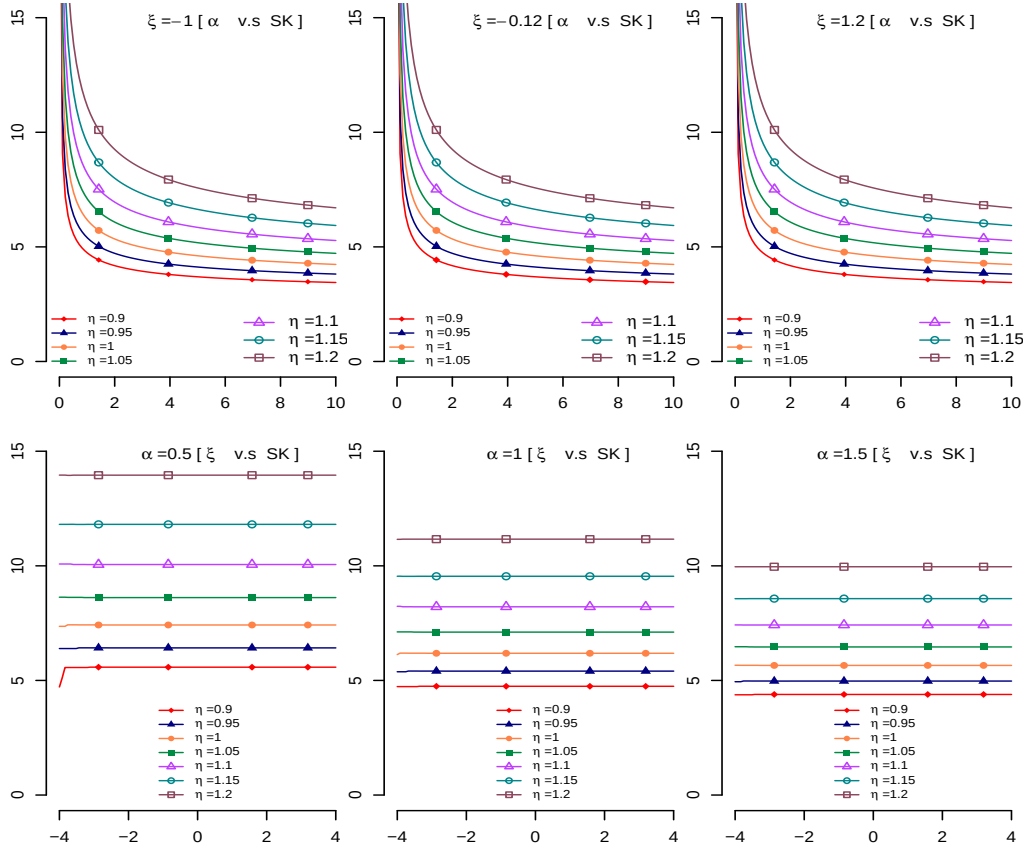


FIGURE 3 Variances of exponentiated lognormal distributions with various values of resilience parameter α , the location and scale parameters of the reference lognormal distribution ξ and η , respectively.

The kernel of the log-likelihood function corresponding to (13) is given by

$$l(\xi, \eta, \alpha) = \alpha \sum_{j=1}^k m_j \ln \Phi \left(\frac{\log(d_j) - \xi}{\eta} \right) + (\alpha - 1) \sum_{i=1}^{n_0} \ln \Phi \left(\frac{\log(x_i) - \xi}{\eta} \right) + n_0 (\log \alpha - \ln \eta) - \frac{1}{2} \sum_{i=1}^{n_0} \left(\frac{\log(x_i) - \xi}{\eta} \right)^2. \quad (14)$$

The maximum likelihood estimator of (ξ, η, α) , denoted by $(\hat{\xi}, \hat{\eta}, \hat{\alpha})$, is the solution to the following optimization problem

$$(\hat{\xi}, \hat{\eta}, \hat{\alpha}) = \sup_{\xi, \eta, \alpha} l(\xi, \eta, \alpha). \quad (15)$$

To derive the score equations and the Hessian matrix for the numerical solution. We need to simplify the notations. Denote, for $j = 1, 2, \dots, k$ and $i = 1, 2, \dots, n_0$

$$\gamma_j = \frac{\log(d_j) - \xi}{\eta} \text{ and } z_i = \frac{\log(x_i) - \xi}{\eta}. \quad (16)$$

With the above notations, the kernel of the log-likelihood function (14) can be re-expressed in the following

$$l(\xi, \eta, \alpha) = \alpha \sum_{j=1}^k m_j \ln \Phi(\gamma_j) + (\alpha - 1) \sum_{i=1}^{n_0} \ln \Phi(z_i) + n_0 (\log \alpha - \ln \eta) - \frac{1}{2} \sum_{i=1}^{n_0} z_i^2. \quad (17)$$

Since γ_j and z_i are functions of ξ and η , the partial derivatives of γ_j and z_i are given by

$$\frac{\partial \gamma_j}{\partial \xi} = \frac{\partial z_j}{\partial \xi} = -\frac{1}{\eta}, \quad \frac{\partial \gamma_j}{\partial \eta} = -\frac{\log(d_j) - \xi}{\eta^2} = -\frac{\gamma_j}{\eta}, \quad \text{and} \quad \frac{\partial z_i}{\partial \eta} = -\frac{\log(x_i) - \xi}{\eta^2} = -\frac{z_i}{\eta}. \quad (18)$$

Note also that, for $t = [\log(s) - \xi]/\eta$,

$$\frac{\partial \Phi(t)}{\partial \xi} = -\frac{\phi(t)}{\eta}, \quad \frac{\partial \Phi(t)}{\partial \eta} = -\frac{t\phi(t)}{\eta}, \quad \frac{\partial \phi(t)}{\partial \xi} = \frac{t\phi(t)}{\eta}, \quad \text{and} \quad \frac{\partial \phi(t)}{\partial \eta} = \frac{t^2\phi(t)}{\eta}. \quad (19)$$

Therefore, the system of score equations is given in the following

$$\frac{\partial l(\xi, \eta, \alpha)}{\partial \xi} = -\frac{1}{\eta} \left[\sum_{j=1}^k \frac{\alpha m_j \phi(\gamma_j)}{\Phi(\gamma_j)} + \sum_{i=1}^{n_0} \frac{(\alpha - 1)\phi(z_i)}{\Phi(z_i)} - \sum_{i=1}^{n_0} z_i \right] = 0, \quad (20)$$

$$\frac{\partial l(\xi, \eta, \alpha)}{\partial \eta} = -\frac{1}{\eta} \left[\sum_{j=1}^k \frac{\alpha m_j \gamma_j \phi(\gamma_j)}{\Phi(\gamma_j)} + \sum_{i=1}^{n_0} \frac{(\alpha - 1)z_i \phi(z_i)}{\Phi(z_i)} + n_0 - \sum_{i=1}^{n_0} z_i^2 \right] = 0, \quad (21)$$

$$\frac{\partial l(\xi, \eta, \alpha)}{\partial \alpha} = \sum_{j=1}^k m_j \ln \Phi(\gamma_j) + \frac{n_0}{\alpha} + \sum_{i=1}^{n_0} \ln \Phi(z_i) = 0. \quad (22)$$

Since the solutions to (20)-(22) have no closed form, numerical methods is used to find the solutions to the score equations. The following Hessian matrix is used the he Newton-Raphson method.

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \xi^2} & \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \xi \partial \eta} & \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \xi \partial \alpha} \\ \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \eta \partial \xi} & \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \eta^2} & \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \eta \partial \alpha} \\ \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \alpha \partial \xi} & \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \alpha \partial \eta} & \frac{\partial^2 l(\xi, \eta, \alpha)}{\partial \alpha^2} \end{pmatrix} \equiv \begin{pmatrix} h_{11}(\xi, \eta, \alpha) & h_{12}(\xi, \eta, \alpha) & h_{13}(\xi, \eta, \alpha) \\ h_{21}(\xi, \eta, \alpha) & h_{22}(\xi, \eta, \alpha) & h_{23}(\xi, \eta, \alpha) \\ h_{31}(\xi, \eta, \alpha) & h_{32}(\xi, \eta, \alpha) & h_{33}(\xi, \eta, \alpha) \end{pmatrix} \quad (23)$$

To derive the individual cells in the Hessian matrix, we will use the following partial derivatives, for $t = [\log(s) - \xi]/\eta$,

$$\frac{\partial}{\partial \xi} \left(\frac{\phi(t)}{\Phi(t)} \right) = \frac{1}{\eta} \frac{\phi(t)}{\Phi(t)} \left(t + \frac{\phi(t)}{\Phi(t)} \right) \quad \text{and} \quad \frac{\partial}{\partial \eta} \left(\frac{\phi(t)}{\Phi(t)} \right) = \frac{t}{\eta} \frac{\phi(t)}{\Phi(t)} \left(t + \frac{\phi(t)}{\Phi(t)} \right) \quad (24)$$

After some algebra, we have

$$\begin{aligned} -h_{11} &= \frac{1}{\eta^2} \left[\sum_{j=1}^k \frac{\alpha m_j \phi(\gamma_j)}{\Phi(\gamma_j)} \left(\gamma_j + \frac{\phi(\gamma_j)}{\Phi(\gamma_j)} \right) + \sum_{i=1}^{n_0} \frac{(\alpha - 1)\phi(z_i)}{\Phi(z_i)} \left(z_i + \frac{\phi(z_i)}{\Phi(z_i)} \right) + n_0 \right], \\ -h_{12} &= -h_{21} = \frac{1}{\eta^2} \left[\sum_{j=1}^k \frac{\alpha m_j \phi(\gamma_j)}{\Phi(\gamma_j)} \left(-1 + \gamma_j^2 + \frac{\gamma_j \phi(\gamma_j)}{\Phi(\gamma_j)} \right) + \sum_{i=1}^{n_0} \frac{(\alpha - 1)\phi(z_i)}{\Phi(z_i)} \left(-1 + z_i^2 + \frac{z_i \phi(z_i)}{\Phi(z_i)} \right) + 2 \sum_{i=1}^{n_0} z_i \right], \\ -h_{13} &= -h_{31} = \frac{1}{\eta} \left[\sum_{j=1}^k \frac{m_j \phi(\gamma_j)}{\Phi(\gamma_j)} + \sum_{i=1}^{n_0} \frac{\phi(z_i)}{\Phi(z_i)} \right], \\ -h_{22} &= \frac{1}{\eta^2} \left[\sum_{j=1}^k \frac{\alpha m_j \gamma_j \phi(\gamma_j)}{\Phi(\gamma_j)} \left(-2 + \gamma_j^2 + \frac{\gamma_j \phi(\gamma_j)}{\Phi(\gamma_j)} \right) + \sum_{i=1}^{n_0} \frac{(\alpha - 1)z_i \phi(z_i)}{\Phi(z_i)} \left(-2 + z_i^2 + \frac{z_i \phi(z_i)}{\Phi(z_i)} \right) + 3 \sum_{i=1}^{n_0} z_i^2 - n_0 \right], \\ -h_{23} &= -h_{32} = \frac{1}{\eta} \left[\sum_{j=1}^k \frac{m_j \gamma_j \phi(\gamma_j)}{\Phi(\gamma_j)} + \sum_{i=1}^{n_0} \frac{z_i \phi(z_i)}{\Phi(z_i)} \right], \\ -h_{33} &= \frac{n_0}{\alpha^2} \end{aligned}$$

Let $n = n_0 + \sum_{i=1}^k m_i$. As the Fisher information matrix is $\mathbf{I} = -E[\mathbf{H}(\xi, \eta, \alpha)]$. The observed Fisher information matrix is given by

$$\widehat{\mathbf{I}}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) = - \begin{pmatrix} h_{11}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) & h_{12}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) & h_{13}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) \\ h_{21}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) & h_{22}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) & h_{23}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) \\ h_{31}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) & h_{32}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) & h_{33}(\hat{\xi}, \hat{\eta}, \hat{\alpha}) \end{pmatrix} \quad (25)$$

The estimated covariance matrix of $(\hat{\xi}, \hat{\eta}, \hat{\alpha})$ is explicitly given by

$$V \equiv \text{cov} \begin{pmatrix} \hat{\xi} \\ \hat{\eta} \\ \hat{\alpha} \end{pmatrix} = \mathbf{I}^{-1}(\xi, \eta, \alpha). \quad (26)$$

The standard large sample theory yields the following asymptotic multivariate normal distribution

$$(\hat{\xi} - \xi_0, \hat{\eta} - \eta_0, \hat{\alpha} - \alpha_0)^T \sim \mathbf{N}(\mathbf{0}, V) \quad (27)$$

where ξ_0 , η_0 and α_0 are the true values of the population mean and the standard deviation.

4 | ASYMPTOTIC CONFIDENCE INTERVALS OF DISTRIBUTION CHARACTERISTICS

4.1 | Mean and Variance

It is common in environmental and related fields that the measurements of interest are asymmetrically distributed. We are interested in making inferences about the mean and variance of the underlying population. Recall that

$$\mu = \int_0^1 \alpha t^{\alpha-1} \exp[\xi + \eta \Phi^{-1}(t)] dt, \quad (28)$$

and

$$\sigma^2 = \int_0^1 \alpha t^{\alpha-1} \exp[2\xi + 2\eta \Phi^{-1}(t)] dt - \mu^2. \quad (29)$$

Using plug-in principle of MLE, we have MLE of population mean and variance as follows

$$\hat{\mu} = \int_0^1 \hat{\alpha} t^{\hat{\alpha}-1} \exp[\hat{\xi} + \hat{\eta} \Phi^{-1}(t)] dt, \quad (30)$$

$$\hat{\sigma}^2 = \int_0^1 \hat{\alpha} t^{\hat{\alpha}-1} \exp[2\hat{\xi} + 2\hat{\eta} \Phi^{-1}(t)] dt - \hat{\mu}^2. \quad (31)$$

Next we establish asymptotic confidence intervals for both population mean μ and variance σ^2 . To construct the confidence interval for the mean, we need following partial derivatives:

$$\begin{aligned}
\mu_\xi &= \frac{\partial \mu}{\partial \xi} = \int_0^1 \alpha t^{\alpha-1} \exp[\xi + \eta \Phi^{-1}(t)] dt, \\
\mu_\eta &= \frac{\partial \mu}{\partial \eta} = \int_0^1 \alpha t^{\alpha-1} \Phi^{-1}(t) \exp[\xi + \eta \Phi^{-1}(t)] dt, \\
\mu_\alpha &= \frac{\partial \mu}{\partial \alpha} = \int_0^1 (t + \alpha^2 - \alpha) t^{\alpha-2} \exp[\xi + \eta \Phi^{-1}(t)] dt, \\
\sigma_\xi^2 &= \frac{\partial \sigma^2}{\partial \xi} = 2 \int_0^1 \alpha t^{\alpha-1} \exp[2\xi + 2\eta \Phi^{-1}(t)] dt - 2\mu\mu_\xi, \\
\sigma_\eta^2 &= \frac{\partial \sigma^2}{\partial \eta} = 2 \int_0^1 \alpha t^{\alpha-1} \Phi^{-1}(t) \exp[2\xi + 2\eta \Phi^{-1}(t)] dt - 2\mu\mu_\eta, \\
\sigma_\alpha^2 &= \frac{\partial \sigma^2}{\partial \alpha} = \int_0^1 (t + \alpha^2 - \alpha) t^{\alpha-2} \exp[2\xi + 2\eta \Phi^{-1}(t)] dt - 2\mu\mu_\alpha.
\end{aligned}$$

We next linearize the MLEs of the mean and variance using a first-order Taylor expansion around the true parameter values $(\xi_0, \eta_0, \alpha_0)$. Let $\mu_{\xi_0}, \mu_{\eta_0}, \mu_{\alpha_0}$ denote the true values of $\mu_\xi, \mu_\eta, \mu_\alpha$, and let $\sigma_{\xi_0}^2, \sigma_{\eta_0}^2, \sigma_{\alpha_0}^2$ denote the true values of $\sigma_\xi^2, \sigma_\eta^2, \sigma_\alpha^2$, respectively. The resulting Taylor expansions of $\hat{\mu}$ and $\hat{\sigma}^2$ are given by

$$\hat{\mu} = \mu_0 + \mu_{\xi_0}(\hat{\xi} - \xi_0) + \mu_{\eta_0}(\hat{\eta} - \eta_0) + \mu_{\alpha_0}(\hat{\alpha} - \alpha_0) + o_p(n^{-1/2}), \quad (32)$$

$$\hat{\sigma}^2 = \sigma_0^2 + \sigma_{\xi_0}^2(\hat{\xi} - \xi_0) + \sigma_{\eta_0}^2(\hat{\eta} - \eta_0) + \sigma_{\alpha_0}^2(\hat{\alpha} - \alpha_0) + o_p(n^{-1/2}). \quad (33)$$

Define

$$D_0 = \begin{pmatrix} \mu_{\xi_0} & \mu_{\eta_0} & \mu_{\alpha_0} \\ \sigma_{\xi_0}^2 & \sigma_{\eta_0}^2 & \sigma_{\alpha_0}^2 \end{pmatrix}. \quad (34)$$

With the notations defined above, we summarize the asymptotic distribution of $(\hat{\mu}, \hat{\sigma}^2)^T$ in the following theorem.

Theorem 3. When the sample size is large, we have

$$(\hat{\mu} - \mu_0, \hat{\sigma}^2 - \sigma_0^2)^T \rightarrow \mathbb{N}_3(\mathbf{0}, \Sigma_0) \quad (35)$$

where $\Sigma_0 = D_0 V_0 D_0^T$, V is specified in (26) and $V_0 = \mathbf{I}^{-1}(\xi_0, \eta_0, \alpha_0)$.

In practical applications, D_0 is replaced by its MLE. That is, $\hat{D}_0$ is obtained by substituting $(\hat{\xi}, \hat{\eta}, \hat{\alpha})$ for $(\xi_0, \eta_0, \alpha_0)$. The MLE of Σ is then given by $\hat{\Sigma} = \hat{D}_0 \hat{V} \hat{D}_0^T$, where V is specified in (26). Let $\Sigma_d = (\sigma_{11}, \sigma_{22})$ denote the main diagonal elements of Σ , and let $\hat{\Sigma}_d = (\hat{\sigma}_{11}, \hat{\sigma}_{22})$ denote the main diagonal elements of $\hat{\Sigma}$.

Corollary 2. When sample size is large, the $100(1 - \theta)\%$ two sided confidence intervals for μ and σ^2 are given, respectively, by

$$\hat{\mu} \pm Z_{1-\theta/2} \sqrt{\hat{\sigma}_{11}} \quad \text{and} \quad \hat{\sigma}^2 \pm Z_{1-\theta/2} \sqrt{\hat{\sigma}_{22}}$$

where $Z_{1-\theta/2}$ is the normal critical value at confidence level $100(1 - \theta)\%$.

For the special case where $\alpha = 1$, i.e. $X \sim LN(\xi, \eta)$, our Taylor expansion will be

$$\begin{aligned}
\hat{\mu} &= \mu_0 + \mu_{\xi_0}(\hat{\xi} - \xi_0) + \mu_{\eta_0}(\hat{\eta} - \eta_0) + o_p(n^{-1/2}) \\
\hat{\sigma}^2 &= \sigma_0^2 + \sigma_{\xi_0}^2(\hat{\xi} - \xi_0) + \sigma_{\eta_0}^2(\hat{\eta} - \eta_0) + o_p(n^{-1/2})
\end{aligned}$$

In this case, under the definition of

$$D_0 = \begin{pmatrix} \mu_{\xi_0} & \mu_{\eta_0} \\ \sigma_{\xi_0}^2 & \sigma_{\eta_0}^2 \end{pmatrix}, \quad (36)$$

the aforementioned asymptotic properties still hold.

The confidence intervals described above will be illustrated with real-world data in Section 7. We now introduce inference for process quantile estimation, which plays a key role in environmental process control and monitoring.

4.2 | Quantile

Estimation of quantiles (also referred to as value-at-risk) is also important in environmental processes such as cleanliness, temperature, relative humidity, pressurization, vibration and acoustic isolation, exposure limits, among others. Let X_q denote the q -th quantile of interest, where the tail probability q is typically large and close to 1. Using the cumulative distribution function (CDF) of the exponentiated lognormal (ELN) distribution given in (1), we have

$$q = \left[\Phi \left(\frac{\log(X_q) - \xi}{\eta} \right) \right]^\alpha.$$

Solve X_q from the above equation, we obtain the q -th quantile as follows

$$X_q = \exp [\eta \Phi^{-1}(q^{1/\alpha}) + \xi] \quad (37)$$

Plugging the MLE of (ξ, η, α) into the above quantile, we have MLE of $\hat{X}_q$. To construct the confidence interval for X_q , need following partial derivatives.

$$\begin{aligned} X_q^\xi &= \frac{\partial X_q}{\partial \xi} = \exp [\eta \Phi^{-1}(q^{1/\alpha}) + \xi] = X_q, \\ X_q^\eta &= \frac{\partial X_q}{\partial \eta} = \exp [\eta \Phi^{-1}(q^{1/\alpha}) + \xi] \Phi^{-1}(q^{1/\alpha}) = \Phi^{-1}(q^{1/\alpha}) X_q, \\ X_q^\alpha &= \frac{\partial X_q}{\partial \alpha} = \exp [\eta \Phi^{-1}(q^{1/\alpha}) + \xi] \frac{-\eta q^{1/\alpha} \log q}{\alpha^2 \phi [\Phi^{-1}(q^{1/\alpha})]} = -\frac{\eta q^{1/\alpha} \log q}{\alpha^2 \phi [\Phi^{-1}(q^{1/\alpha})]} X_q. \end{aligned}$$

Let $D_{xq} = (X_q^\xi, X_q^\eta, X_q^\alpha)$. Let X_q^0 and D_{xq}^0 be the true value q -th quantile and vector D_{xq}^0 , respectively. Then

$$\hat{X}_q = X_q + D_{xq}^0 (\hat{\xi} - \xi_0, \hat{\eta} - \eta_0, \hat{\sigma}^2 - \sigma_0^2)^T + o_p(n^{-1/2}).$$

The asymptotic distribution of the MLE of $\hat{X}_q$ and confidence interval are given in the following theorem.

Theorem 4. When the sample size is large, we have

$$\hat{X}_q - X_q \rightarrow \mathbb{N}(0, D_{xq} V D_{xq}^T) \quad (38)$$

Using the MLE proposed before for V and D_{xq} , denoted by $\hat{V}$ and $\hat{D}_{xq}$, before, we can construct the $100(1 - \theta)\%$ one-sided asymptotic confidence interval for q -th quantile X_q as follows

$$\hat{X}_q + Z_{1-\theta} \sqrt{\hat{D}_{xq} \hat{V} \hat{D}_{xq}^T} \quad (39)$$

where Z_q is the normal critical value at confidence level $100(1 - \theta)\%$.

The lower quantile is rarely used in the environmental study. However, the confidence interval for lower quantile can be similarly constructed with form $\hat{X}_{1-\theta} - Z_{1-\theta} \sqrt{\hat{D}_{xq} \hat{V} \hat{D}_{xq}^T}$, where θ is close to 0.

5 | MODEL EVALUATION

In this section, we assess model performance from two perspectives: goodness-of-fit for evaluating correct model specification, and significance testing of the resilience parameter α to guard against under-fitting and over-fitting in practical applications.

5.1 | Goodness-of-fit Test

One practical question that must be addressed is how well the model fits the data. For complete data, numerous testing procedures have been proposed in the literature. Some tests are also available for the case of right random Type II censored and left-truncated data. In environmental studies, censored data arise due to limitations in measurement accuracy and precision, specifically when observations fall below fixed detection limits of different measuring instruments. In this paper, we adopt the Pearson χ^2 goodness-of-fit test to assess the fit of the exponentiated lognormal distribution.

The Pearson χ^2 test is defined based on a comparison between the observed frequencies and the expected (theoretical) frequencies across bins (data windows). The approach to defining bins may vary. We propose the following algorithm to specify the bins and determine the corresponding observed and expected frequencies for use in the Pearson χ^2 goodness-of-fit test.

Algorithm: Assume that $d_1 < d_2 < \dots < d_k$ are the k censoring limits ordered in ascending order, and the number of observations that are bigger than d_k is large.

1. Let $d_i = X_{qi}, i = 1, 2, \dots, k$. Insert the following partition points into the data

$$X_{q1} < X_{q2} < \dots < X_{qk} < X_{q(k+1)} < \dots < X_{qp}$$

to divide the interval $(0, +\infty)$ into $p + 1$ sub-intervals

$$(0, X_{q1}], (X_{q1}, X_{q2}], \dots, (X_{q(p-1)}, X_{qp}], (X_{qp}, +\infty)$$

such that each sub-interval contains more than five observations.

2. Compute the probability that a sample point falls into the aforementioned intervals based on population $ELN(\hat{\xi}, \hat{\eta}, \hat{\alpha})$, and the corresponding theoretical (expected) frequencies based on the sample size n , denoted as $E_0, E_1, \dots, E_p$.
3. For $1 \leq i \leq n$, let $O_0 = \#\{x_i \leq X_{q1}\}$, $O_j = \#\{X_{qj} < x_i \leq X_{q(j+1)}\}$ for $j = 1, 2, \dots, p - 1$ and $O_p = \#\{x_i > X_{qp}\}$ be observed frequencies of the sample points in each sub-interval.
4. Conduct Pearson χ^2 test theoretical and observed frequencies.

The above algorithm guarantees that the minimum frequency in each partition of the observed data is greater than 5. If the frequency in any partition of the generated data is less than 5, partitions should be combined so that the combined frequency exceeds 5. Note also that the partitions are not of equal width.

Theorem 5. Assume that the number of observations bigger than the maximum detection limit. Using above algorithm to find the observed and expected frequencies O_j and E_j for $j = 0, 1, \dots, p$, we have

$$T = \sum_{j=0}^p \frac{(O_j - E_j)^2}{E_j} \sim \chi^2(p - 3) \quad (40)$$

under the null hypothesis that the exponentiated lognormal distribution fits the data well.

Based on the theorem above, we can perform the Pearson χ^2 goodness-of-fit test. If the p -value of the test is less than the chosen significance level, the null hypothesis should be rejected.

5.2 | Detection of Potential Over-Fitting and Under-Fitting

Parsimony is a desirable feature in statistical modeling. Since the lognormal distribution is a special case of the exponentiated lognormal distribution, it is necessary to develop a procedure to detect potential over-fitting and under-fitting. In other words, we can test

$$H_0 : \alpha = 1 \quad vs \quad H_1 : \alpha \neq 1. \quad (41)$$

There are different ways to test this hypothesis. For example, one can equivalently test whether the area under the ROC curve given in (5) equals 0.5. In this paper, we present two statistical testing methods: the likelihood ratio test (LRT) and the Wald-type test.

Theorem 6. Assume that the sample size is large and the underlying distribution is exponentiated lognormal. We have

$$L = -2 \ln \left[\frac{L(\hat{\theta}_r; x)}{L(\hat{\theta}_F; x)} \right] = -2 [l(\hat{\theta}_r; x) - l(\hat{\theta}_F)] \sim \chi^2(1), \quad (42)$$

where $\hat{\theta}_F = (\hat{\xi}, \hat{\eta}, \hat{\alpha})$ is the full MLE of (17), and $\hat{\theta}_R = (\hat{\xi}, \hat{\eta})$ is the restricted MLE under the constraint $\alpha = 1$.

Theorem 7. Assume that the sample size is large and the underlying distribution is exponentiated lognormal. We have

$$W = \frac{(\hat{\alpha} - 1)^2}{\hat{v}_{33}} \sim \chi^2(1), \quad (43)$$

where $\hat{v}_{33}$ is the third diagonal element of the MLE of V given in (26).

If the observed p -value based on (42) is less than the given significance level, we reject the null hypothesis that the underlying distribution is exponentiated lognormal. Otherwise, we should assume lognormal distribution. The Wald-type test (43) based on (27) produces the same conclusion.

6 | SIMULATION STUDIES

In this section, we will provide bootstrap confidence intervals as a supplement to the asymptotic ones, and perform some simulation studies focusing on the following two aspects: the performance of asymptotic confidence intervals for model parameters $(\hat{\xi}, \hat{\eta}, \hat{\alpha})$ based on (27) as well as the performance of the LRT (42) and the Wald-Type Test (43) under the testing problem (45).

6.1 | Bootstrap Confidence Interval

As both sample size and censoring proportion can impact the asymptotic confidence intervals in previous sections, we propose the percentile bootstrap confidence interval based on the method of moments as a supplement.

The bootstrap samples are generated from the population $ELN(\hat{\xi}, \hat{\eta}, \hat{\alpha})$, and we can use the inverse CDF function method to derive the random numbers which are then used to form the bootstrap samples. For a given random number U from uniform population $U[0, 1]$, the inverse of the CDF (1) that has the following form

$$X = \exp [\xi + \eta \Phi^{-1}(U^{1/\alpha})] \quad (44)$$

is the random number from the exponentiated log-normal population of (1). Based on this, the bootstrap confidence interval for the parameter γ generated through the following algorithm.

Algorithm:

1. Compute the MLEs $(\hat{\xi}, \hat{\eta}, \hat{\alpha})$ from the real-world data by solving optimization problem

$$\max_{\xi, \eta, \alpha} (\xi, \eta, \alpha)$$

2. Generate the following B groups of bootstrap samples from population $ELN(\hat{\xi}, \hat{\eta}, \hat{\alpha})$

$$\left\{ x_1^{*(i)}, x_2^{*(i)}, \dots, x_n^{*(i)} \right\}, \quad i = 1, 2, \dots, B$$

3. For each bootstrap sample, compute the method of moments estimator of the parameter of interest γ , yielding

$$\hat{\gamma}^{*(1)}, \hat{\gamma}^{*(2)}, \dots, \hat{\gamma}^{*(B)}$$

4. Construct the θ -level percentile confidence interval, for example, $(\hat{\gamma}_\theta^*, \hat{\gamma}_{1-\theta}^*)$.

In the subsequent simulation experiments, we will continue to employ the inverse CDF method for generating random numbers. We focus on simulation experiments under the single censoring limit scenario. In this case, the experimental data can be obtained by simply replacing any simulated data point that falls below the censoring limit with the value of the limit itself.

6.2 | Coverage probability

To evaluate the finite-sample performance of the maximum likelihood estimators for the parameters of the exponentiated log-normal (ELN) distribution, we conducted a Monte Carlo simulation study. The primary objective was to assess the coverage probabilities of asymptotic confidence intervals for the model parameters—namely the location parameter ξ , scale parameter η , and resilience parameter α under various conditions. Specifically, we examined censoring proportion and true parameter values on the accuracy. Data were generated from the ELN distribution under the left-censored schemes, with censoring thresholds set at fixed detection limits to reflect common practices in environmental monitoring.

In this simulation study, two parameter configurations are considered: $(\xi, \eta, \alpha) = (1, 0.5, 0.2)$ and $(1, 0.5, 1)$. The nominal confidence level is set to 0.95, and three censoring proportions are examined: 1%, 10%, and 20%. The results are summarized in Figure 4 .

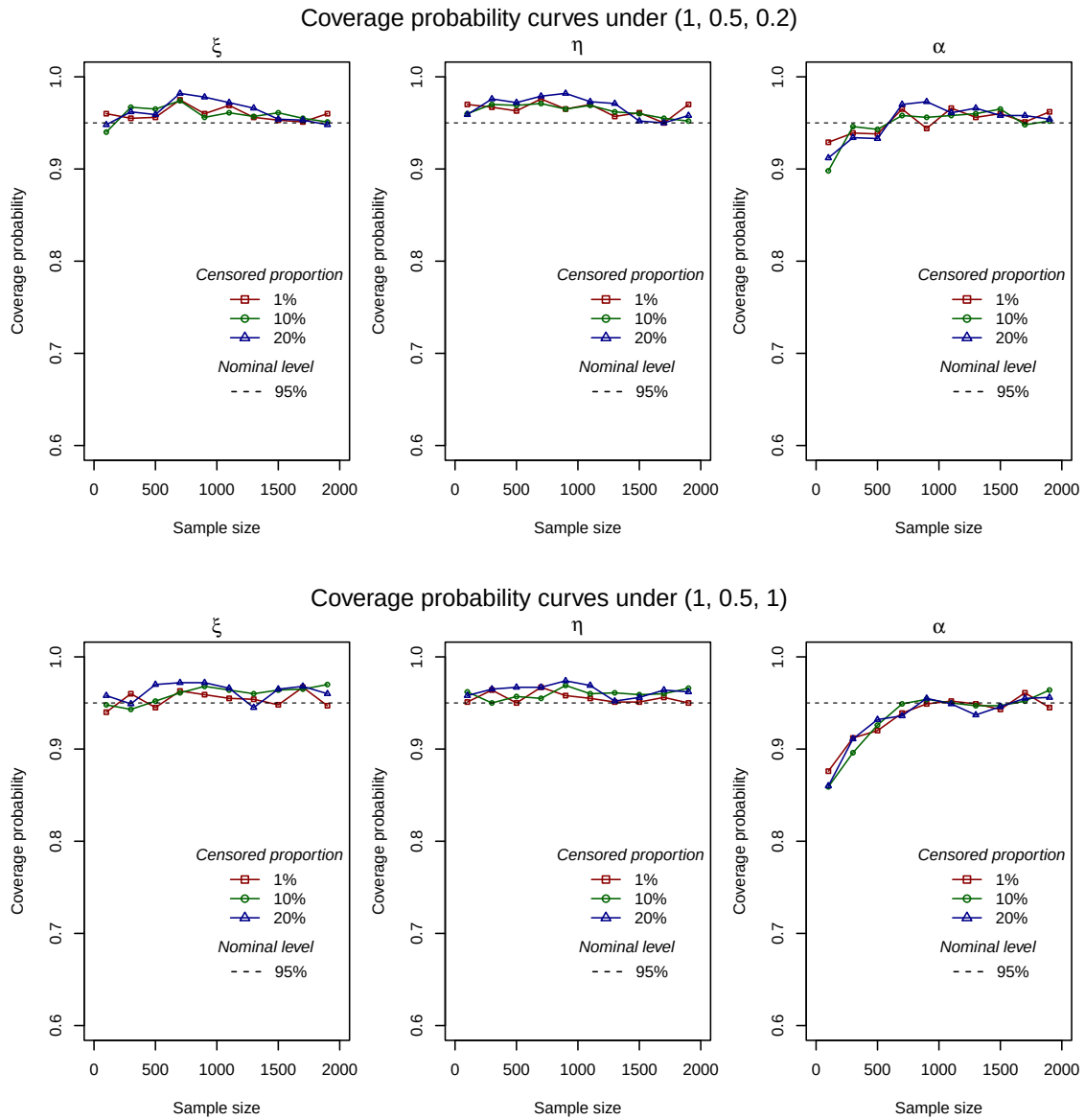


FIGURE 4 Coverage Probability Curves for the Model Parameters (ξ, η, α) at different censoring proportions.

Figure 4 indicates that, The coverage probabilities of the confidence intervals for ξ and η are close to the nominal level, with negligible over-coverage that is not significantly affected by sample size or censoring proportion. The coverage probability for the resilience parameter α behaves slightly differently. When the sample size is small, the coverage probability is noticeably lower than the nominal level. However, as the sample size increases, the coverage probabilities for different censoring proportions approach the nominal level and become stable.

6.3 | Power Analysis

Since the exponentiated distribution is reduced to the lognormal distribution if $\alpha = 1$. Therefore, it is practically important to test $\alpha = 1$ to avoid potential over-fitting and under-fitting issue. Here we present a power analysis for testing the following hypothesis

$$H_0 : \alpha = 1 \quad vs \quad H_1 : \alpha \neq 1. \quad (45)$$

We evaluate the power of both the likelihood ratio test (LRT) in (42) and the Wald-type test in (43) proposed in the previous sections. For illustrative purposes, we continue to generate samples under a single censoring limit, using the same censoring proportions as in the coverage probability simulations. The location and scale parameters of the exponentiated lognormal population are fixed at $(\xi, \eta) = (1, 0.5)$. The values of the resilience parameter under the alternative hypothesis are chosen as $(0.2, 0.6, 1.0, 1.4, 1.8, 2.2, 2.6, 3.0, 5.0)$, with sample sizes ranging from 50 to 250. The resulting power curves are presented in Figure 6.3. The resulting power curves are summarized in the following Figure .

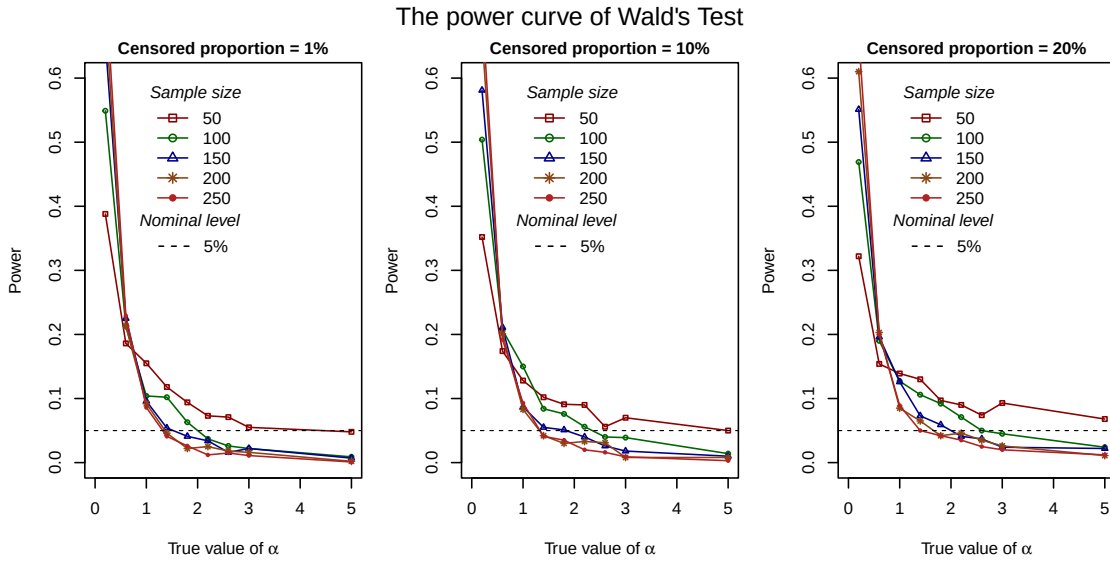
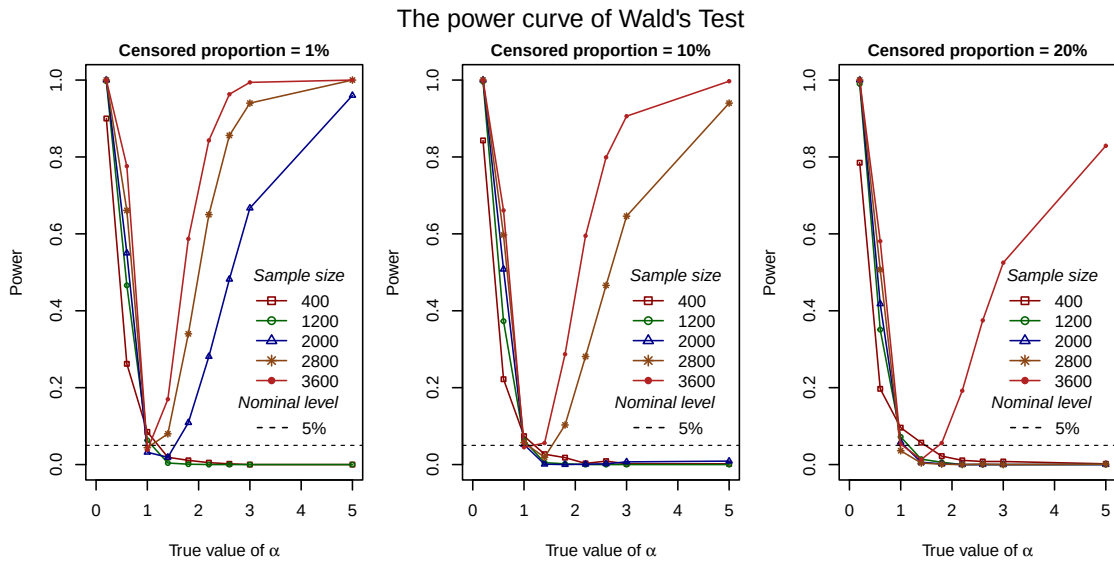
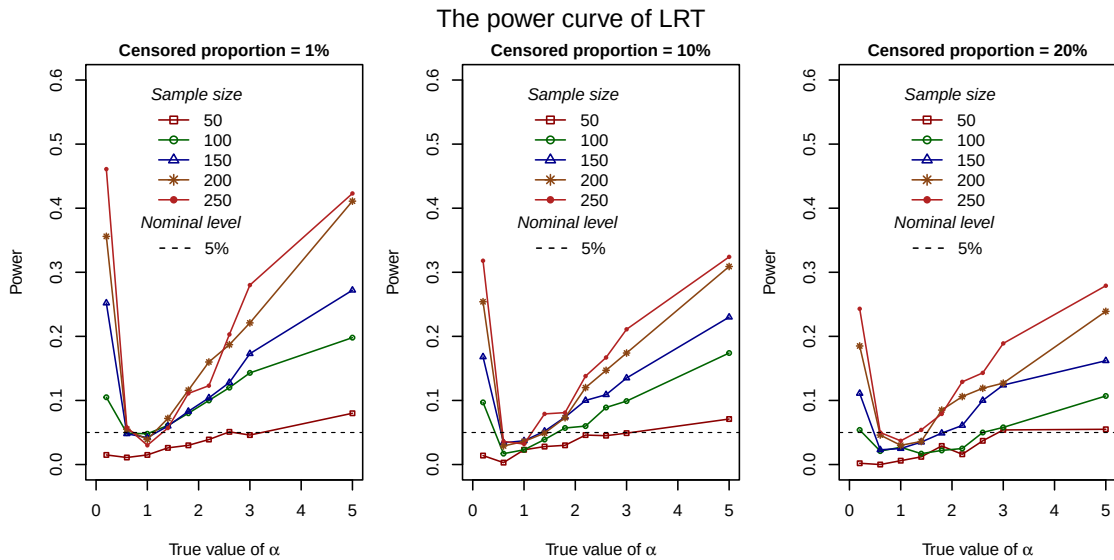


FIGURE 5 Power Curves for small-sample scenarios

Figure 5 and Figure 6 show the power of the Wald test under different sample sizes and censoring proportions. We observe that the Wald test exhibits high power and is less sensitive to censoring proportion and sample size when $\alpha < 1$, largely due to the small effect size. Additionally, for a fixed α , power tends to increase with sample size within each panel and decrease as the censoring proportion increases across panels.

However, when $\alpha > 1$, the test requires a substantially larger sample size to achieve a given level of power, particularly as the censoring proportion increases. This pattern is clear in Figure 6 .

In contrast to the Wald test, the likelihood ratio test (LRT) exhibits lower power when $\alpha < 1$, as evident from a comparison of the power curves in Figures 7 and 6 . Under small sample sizes and higher censoring proportions, the observed significance level of the LRT tends to fall below the nominal level—a pattern opposite to that of the Wald test. Despite this, the LRT demonstrates substantially greater power than the Wald test under identical conditions.

**FIGURE 6** Power Curves for small-sample scenarios**FIGURE 7** Power Curves for large-sample scenarios

In summary, we recommend the LRT for hypothesis testing during practical applications, aligning with the recommendations in the literature.. In particular, when $\alpha < 1$ and the sample size is relatively small, the Wald-type statistic can be used as an alternative test.

7 | REAL-WORLD APPLICATIONS

In this section, we work on three numerical examples that contain single and multiple detection limits based on real-world applications in environmental studies. We will report the result of goodness-of-fit test and the MLE of parameters along with their

asymptotic confidence intervals and percentile bootstrap confidence intervals for population parameters and the exponentiated lognormal population.

7.1 | Water Quality of Giles Creek

Total recoverable chromium concentrations have been monitored in the streamflows of Gales Creek, Oregon, using publicly available data provided by the U.S. Geological Survey (USGS) at <https://waterdata.usgs.gov/monitoring-location/453026123063401/>. The dataset includes 62 records collected between January 2009 and October 2018 at the Gales Creek station near New Highway 47. A key analytical challenge arises from the presence of multiple detection limits: due to advancements in analytical technology and changes in laboratory protocols over the monitoring period, the detection limit for chromium has decreased over time. As a result, earlier observations are left-censored at a higher threshold of 0.6, while more recent samples are censored at a lower, more sensitive limit of 0.4. The objective of this analysis is to estimate the mean recoverable chromium concentration at the station during the data collection period.

By maximizing the log-likelihood function in (17), we obtain the estimates $(\hat{\xi}, \hat{\eta}, \hat{\alpha}) = (-5.4790, 2.1175, 101.6202)$. Using the partition points 0.400, 0.600, 0.729, 0.833, 1.010, 1.290, 1.620, 2.280, the p-value for the goodness-of-fit χ^2 test is 0.1885, suggesting that the data are consistent with an exponentiated lognormal population. The likelihood ratio test of the hypothesis $H_0 : \alpha = 1$ yields a p-value of 0.072. At a significance level of 0.05, we fit the regular lognormal model to the data to avoid overfitting. Asymptotic results for the lognormal parameters and the population mean and variance are summarized in Table 1.

TABLE 1 Asymptotic confidence intervals for the parameters

Parameter	MLE	SE	LCL	UCL	CI
ξ	-0.2109	0.1345	-0.4745	0.0527	$(-0.4745, 0.0527)$
η	1.0089	0.1108	0.7918	1.2260	$(0.7918, 1.2260)$
μ	1.3473	0.2129	0.9300	1.7646	$(0.9300, 1.7646)$
σ^2	3.2080	1.8808	-0.4783	6.8943	$(0, 6.8943)$

We also constructed bootstrap percentile confidence intervals for the population mean and variance using the algorithm detailed in Section 6.1. The resulting percentile bootstrap confidence intervals are $\mu \in (1.2456, 1.4606)$ and $\sigma^2 \in (2.0647, 5.6527)$. The bootstrap confidence intervals demonstrate improved performance over the corresponding asymptotic intervals.

7.2 | Thiamethoxam Concentrations in Pollen

In 2015, the Ministry of the Environment, Conservation and Parks, in partnership with the Ontario Ministry of Agriculture, Food and Rural Affairs, established the Pollen Monitoring Network as part of Ontario's Pollinator Health Action Plan. The initiative was designed to assess changes in pesticide concentrations—specifically neonicotinoid insecticides, other insecticides, and fungicides—following the implementation of restrictions on the use of neonicotinoid-treated corn and soybean seeds. A publicly available dataset from the network provides measurements of pesticide concentrations in pollen samples collected from honeybee yards across Ontario. The raw data are accessible at <https://data.ontario.ca/en/dataset/pollen-monitoring-network-study>.

Among the suite of pesticides monitored by the network, this study focuses specifically on the neonicotinoid insecticide Thiamethoxam. As a primary compound used in the seed treatments targeted by Ontario's 2015 restrictions, its concentration in pollen serves as a key indicator for evaluating the policy's effectiveness. Furthermore, Thiamethoxam is of particular toxicological concern for pollinators; even sub-lethal exposures have been linked to impaired foraging and colony health. Therefore, a detailed statistical analysis of post-regulation Thiamethoxam levels in pollen is essential for directly assessing the residual risks

to honeybee populations in agricultural regions of Ontario. A modified and reorganized version of this subset was built in the R package *NADA2*.

The maximum likelihood estimates of the model parameters are $(\hat{\xi}, \hat{\eta}, \hat{\alpha}) = (0.4851, 1.5925, 0.2045)$. Using the partition points 0.05, 0.06, 0.08, 0.11, 0.14, 0.21, 0.24, 0.27, 0.32, 0.41, 0.48, 0.56, 0.62, 0.82, 1.00, 1.50, 2.20, 2.90, 4.50, 10.00, the p-value for the χ^2 goodness-of-fit test is 0.2948. Since $\hat{\alpha} < 1$ and in light of the discussion in the power analysis section, the Wald-type statistic is employed to test for overfitting, yielding a p-value of 0.0005. Based on the two tests above, the data are indeed consistent with the $ELN(\xi, \eta, \alpha)$ distribution. Table 2 presents the asymptotic confidence intervals for the model parameters.

TABLE 2 Asymptotic confidence intervals for the model parameters

Parameter	MLE	SE	LCL	UCL	CI
ξ	0.4851	1.6327	-2.7149	3.6851	(-2.7149, 3.6851)
η	1.5925	0.6037	0.4094	2.7757	(0.4094, 2.7757)
α	0.2045	0.2286	-0.2434	0.6525	(0, 0.6525)

The asymptotic sampling distribution of the maximum likelihood estimators for the mean and variance is approximated using the first-order delta method. This approach involves evaluating the following improper integral:

$$\mu_{\alpha} = \frac{\partial \mu}{\partial \alpha} = \int_0^1 (t + \alpha^2 - \alpha) t^{\alpha-2} \exp[\xi + \eta \Phi^{-1}(t)] dt,$$

where

$$t = \Phi \left(\frac{\log y - \xi}{\eta} \right).$$

Since 0 is a singular point at the lower limit of integration, the integral is improper. We approximate it using the following method:

$$\mu_{\alpha} = \frac{\partial \mu}{\partial \alpha} \approx \int_{\epsilon}^1 (t + \alpha^2 - \alpha) t^{\alpha-2} \exp[\xi + \eta \Phi^{-1}(t)] dt$$

where a small lower integration limit ϵ is used, provided that the corresponding $y_{\min}$ is sufficiently small (i.e., smaller than the lowest detection limit). Identifying an appropriate confidence interval requires some trial and error. Since there is only one censoring limit $d = 0.05$ in this example, we tested three values of ϵ , corresponding to $y_{\min} = 0.010$, 0.005, and 0.001. The resulting three approximate asymptotic confidence intervals for the mean are summarized in the table below.

TABLE 3 Asymptotic confidence intervals for the mean and variance

Parameter	MLE	$y_{\min}$	SE	LCL	UCL	CI
μ	1.4161	0.010	0.5311	0.3752	2.4571	(0.3752, 2.4571)
		0.005	0.7120	0.0206	2.8116	(0.0206, 2.8116)
		0.001	3.3387	-5.1277	7.9599	(-5.1277, 7.9599)
σ^2	85.0936	0.010	151.5709	-211.9800	382.1672	(0, 382.1672)
		0.005	152.1869	-213.1873	383.3745	(0, 383.3745)
		0.001	158.7805	-226.1105	396.2977	(0, 396.2977)

The percentile bootstrap method yields confidence intervals of (1.0210, 2.0723) for the mean and (12.7895, 375.8480) for the variance. These intervals compare favorably to their asymptotic counterparts, suggesting improved performance.

7.3 | San Joaquin Valley Underground Water

Deverel and Millard (1988) studied the distribution of tracement concentrations in shallow ground-waters from Alluvial Fan Zone and Basin-Trough Zone underneath the San Joaquin Valley, California. The dataset contains multiple detection limits and a few missing values as well. This is a classical data environmental data set analyzed by various researchers for various purposes. Peng (2010) used this data to illustrate the MLE of lognormal parameters. Millard and Deverel (1988) re-analyzed copper and zinc concentrations using nonparametric rank based two sample location tests to compare the copper and zinc concentrations in the two geologic zones. In this example, we use the 68 Zinc concentrations (including one missing value, denoted by ms) given in micrograms per liter:

<10, 9, ms, 5, 18, <10, 12, 10, 11, 11, 19, 8, <3, <10, <10, 10, 10, 10, 10, <10, 10, <10, 10, <10, 10, <10, 10, 20, 20, <10, 20, 20, 20, <10, 10, 20, 620, 40, 50, 33, 10, 20, 10, 10, 10, 30, 20, 10, 20, 20, 20, <10, 20, 23, 17, 10, <10, 10, 20, 29, 20, <10, 10, <10, 10, 7, <10.

In this analysis and thereafter, we will ignore the missing measurements (treat it as if it is missing at completely random, MCR) and remove the obvious abnormal data point of 620. We fit this data to exponentiated lognormal distribution $ELN(\xi, \eta, \alpha)$ to this data.

The MLEs of the model parameters are $(\hat{\xi}, \hat{\eta}, \hat{\alpha}) = (2.3396, 0.6118, 1.2474)$. Using the partition points 3, 10, 18, 20, the χ^2 goodness-of-fit test yields a p-value of 2.2204×10^{-16} , which strongly rejects the hypothesis that the data follow an exponentiated lognormal distribution. Therefore, we do not proceed with any further analysis of this dataset.

References

- Cohen, A. C. (1957). On the solution of estimating equations for truncated and censored samples from normal populations. *Biometrika*, 44, 225–236.
- Cohen, A. C. (1959). Simplified estimators for the normal distribution when samples are singly censored or truncated. *Technometrics*, 1, 217–237.
- Cohen, A. C. (1991). *Truncated and censored samples: Theory and applications* (Vol. 119). New York: Marcel Dekker Inc.
- Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap methods and their application* (Vol. 1). Cambridge: Cambridge University Press.
- Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1), 1–38.
- Deverel, S. J., & Millard, S. P. (1988). Distribution and mobility of selenium and other trace elements in shallow ground water of the western san joaquin valley, california. *Environmental Science & Technology*, 22, 697–702.
- Efron, B. (1979). Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26.
- Efron, B. (1981a). Censored data and the bootstrap. *Journal of the American Statistical Association*, 76(374), 312–319.
- Efron, B. (1981b). Nonparametric estimates of standard error: the jackknife, the bootstrap and other methods. *Biometrika*, 68(3), 589–599.
- Efron, B. (1985). Bootstrap confidence intervals for a class of parametric problems. *Biometrika*, 72(1), 45–58.
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap* (Vol. 57). New York: Chapman and Hall.
- El-Shaarawi, A. H. (1989). Inferences about the mean from censored water quality data. *Water Resources Research*, 25(4), 685–690.
- El-Shaarawi, A. H., & Esterby, S. R. (1992). Replacement of censored observations by a constant: an evaluation. *Water Research*, 26(6), 835–844.
- Fletcher, R. (1970). A new approach to variable metric algorithms. *The Computer Journal*, 13(3), 317–322.
- Fletcher, R., & Reeve, C. M. (1964). Function minimization by conjugate gradients. *The Computer Journal*, 7(2), 149–154.
- Gilliom, R., & Helsel, D. (1986). Estimation of distributional parameters for censored trace level water quality data 1. estimation techniques. *Water Resources Research*, 22(2), 135–146.
- Goldfarb, D. (1970). A family of variable-metric methods derived by variational means. *Mathematics of Computation*, 24,

23–26.

- Gupta, A. K. (1952). Estimation of the mean and standard deviation of a normal population from a censored sample. *Biometrika*, 39, 260–273.
- Helsel, D. (1990). Less than obvious: statistical treatment of data below the detection limit. *Environmental Science & Technology*, 24(12), 1767–1774.
- Helsel, D. (2005). *Nondetects and data analysis: statistics for censored environmental data*. New York: Wiley.
- Helsel, D. (2006). Fabricating data: How substituting values for nondetects can ruin results, and what can be done about it. *Chemosphere*, 65, 2434–2439.
- Helsel, D., & Gilliom, R. (1986). Estimation of distributional parameters for censored trace level water quality data 2. verification and applications. *Water Resources Research*, 22(2), 147–155.
- Marshall, A. W., & Olkin, I. (2007). *Life distributions: Structure of nonparametric, semiparametric, and parametric families*. New York, NY, USA: Springer.
- Martínez-Flórez, G., Bolfarine, H., & Gómez, H. W. (2014). The log-power-normal distribution with application to air pollution. *Environmetrics*, 25(1), 44–56.
- Martinez-Florez, G., Bolfarine, H., & Gomez, H. W. (2014). The log-power-normal distribution with application to air pollution. *Environmetrics*, 25, 44–56.
- Millard, S. P., & Deverel, S. J. (1988). Nonparametric statistical methods for comparing two sites based on data with multiple nondetect limits. *Water Resources Research*, 24(12), 2087–2098.
- Nelder, J. A., & Mead, R. (1965). A simplex method for function minimization. *The Computer Journal*, 7(4), 308–313.
- Peng, C. (2010). Interval estimation of population parameters based on environmental data with detection limits. *Environmetrics*, 21(6), 645–658.
- Pepe, M. S. (2004). *The statistical evaluation of medical tests for classification and prediction*. New York: Oxford University Press.
- Persson, T., & Rootzen, H. (1977). Simple and highly efficient estimators for a type I censored normal sample. *Biometrika*, 64, 123–128.
- Shanno, D. F. (1970). Conditioning of quasi-Newton methods for function minimization. *Mathematics of Computation*, 24, 647–656.
- Shumway, R. H., Azari, A. S., & Johnson, P. (1989). Estimating mean concentration under transformation for environmental data with detection limits. *Technometrics*, 31, 347–356.
- Singh, A., & Nocerino, J. (2002). Robust estimation of mean and variance using environmental data sets with below detection limit observations. *Chemometrics and Intelligent Laboratory Systems*, 60(1), 69–86.
- Tiku, M. L. (1967). Estimating the mean and standard deviation from a censored normal sample. *Biometrika*, 54, 155–165.
- U.S. Environmental Protection Agency. (1992). *Statistical analysis of ground water monitoring data at rare facilities, addendum to interim final guidance* (Tech. Rep.). Office of Solid Waste, Waste Management Division.
- Zhao, Y., & Frey, H. C. (2006). Uncertainty for data with non-detects: air toxic emissions from combustion. *Human and Ecological Risk Assessment*, 12, 1171–1191.

